

Effect of pistachio diet on lipid parameters, endothelial function, inflammation, and oxidative status: a prospective study.



OBJECTIVE: Recent studies have suggested that nuts have favorable effects beyond lipid lowering. We aimed to investigate effect of the Antep pistachio (*Pistacia vera* L.) on blood glucose, lipid parameters, endothelial function, inflammation, and oxidation in healthy young men living in a controlled environment. **METHODS:** A Mediterranean diet was administered to normolipidemic 32 healthy young men (mean age 22 y, range 21-24) for 4 wk. After 4 wk, participants continued to receive the Mediterranean diet but pistachio was added for 4 wk by replacing the monounsaturated fat content constituting approximately 20% of daily caloric intake. Fasting blood samples and brachial endothelial function measurements were performed at baseline and after each diet. **RESULTS:** Compared with the Mediterranean diet, the pistachio diet decreased glucose ($P<0.001$, $-8.8\pm 8.5\%$), low-density lipoprotein ($P<0.001$, $-23.2\pm 11.9\%$), total cholesterol ($P<0.001$, $-21.2\pm 9.9\%$), and triacylglycerol ($P=0.008$, $-13.8\pm 33.8\%$) significantly and high-density lipoprotein ($P=0.069$, $-3.1\pm 11.7\%$) non-significantly. Total cholesterol/high-density lipoprotein and low-density lipoprotein/high-density lipoprotein ratios decreased significantly ($P<0.001$ for both). The pistachio diet significantly improved endothelium-dependent vasodilation ($P=0.002$, 30% relative increase), decreased serum interleukin-6, total oxidant status, lipid hydroperoxide, and malondialdehyde and increased superoxide dismutase ($P<0.001$ for all), whereas there was no significant change in C-reactive protein and tumor necrosis factor- α levels. **CONCLUSION:** In this trial, we demonstrated that a pistachio diet improved blood glucose level, endothelial function, and some indices of inflammation and oxidative status in healthy young men. These findings are in accordance with the idea that nuts, in particular pistachio nuts, have favorable effects beyond lipid lowering that deserve to be evaluated with prospective follow-up studies. Copyright 2010. Published by Elsevier Inc.